Adaptive Stress Testing for Algorithmic Trading: An Importance Sampling Approach

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***Abstract*—Validating the safety of algorithmic trading strategies is challenging because catastrophic failure events, such as large, rapid drawdowns are rare under normal market conditions, making naïve Monte Carlo (MC) sampling computationally expensive. We apply Adaptive Stress Testing (AST), a reinforcement learning framework that trains an adversarial agent to efficiently discover worst-case failure trajectories in a target system. We evaluate AST against a Simple Moving Average (SMA) crossover trading strategy in a simulated market environment across two victim configurations: a baseline strategy (V1, with 5-step exit lag) and a more robust strategy (V2, 3-step exit lag). An adversarial Proximal Policy Optimization (PPO) agent is trained against each victim, and importance sampling (IS) is used to translate the biased adversarial failure trajectories into unbiased estimates of the real-world failure probability under the nominal market distribution. Our results (mean $\pm$ std over 5 evaluation trials) show that AST achieves $100.0 \pm 0.0\%$ failure rate against V1 compared to only $0.4 \pm 0.3\%$ for MC, a $\geq 250\times$ improvement in failure discovery, while the IS estimate reveals these failures have probability $\approx 1.5\times10^{-11}$ under natural market noise. Against V2, an adversary trained with twice the compute budget (400k vs. 200k timesteps) achieves only $62.0 \pm 2.6\%$ failure, with IS-estimated probability 2.0×10^{-3}, more than eight orders of magnitude higher, revealing that V2's failures, though harder to find, are far more realistic. These results expose a critical insight, namely that raw failure rate is a deceptive safety metric. V1 appears more dangerous by failure rate (100.0% vs. 62.0%), yet IS reveals V2's failures are more than eight orders of magnitude more likely under natural market conditions. The combination of adversarial sample complexity and IS-corrected failure probability, not merely failure rate, provides the full picture of safety.**

***Index Terms*—adaptive stress testing, algorithmic trading, safety validation, reinforcement learning, importance sampling, failure probability estimation**

I. Introduction

The deployment of automated algorithmic trading systems in real financial markets raises fundamental safety questions: Under what market conditions might the system suffer a catastrophic loss, and how likely are such conditions to occur? Answering these questions is difficult because catastrophic failure events, such as a portfolio losing a majority of its value within minutes, are rare under typical market dynamics. Traditional validation approaches based on Monte Carlo simulation are statistically inefficient for rare-event estimation, requiring an exponentially large number of samples to reliably observe even a single failure.

Adaptive Stress Testing (AST) [1]–[3] addresses this inefficiency by framing the failure-finding problem as a Markov Decision Process (MDP) in which a learned adversarial agent controls the stochastic inputs to the system under test. The adversary is trained via reinforcement learning (RL) to maximize the probability of inducing failures, effectively acting as an importance-weighted sampler over the space of failure-inducing trajectories.

In this work, we apply AST to validate two versions of a simulated SMA crossover trading strategy, which represents a widely deployed class of technical analysis algorithms. We make the following implementation contributions:

1) We construct a Gymnasium simulation environment that models an adversarial stochastic market and two victim trading strategy variants (V1 and V2, with 5-step and 3-step exit lags, respectively, where the exit lag represents the number of consecutive bearish steps required before the position is liquidated).
2) We train PPO-based adversaries against each victim and compare failure discovery efficiency against a Monte Carlo baseline.
3) We implement importance sampling to estimate the real-world probability of failure under the nominal $\mathcal{N}(0, 1)$ market distribution.
4) We demonstrate that iterative adversarial robustness improvement (reducing the exit lag from 5 to 3 steps between V1 and V2) substantially increases adversarial sample complexity: against V2, even doubling the training budget (400k vs. 200k timesteps) yields only $62.0 \pm 2.6\%$ failure rate versus 100.0% for V1, while IS reveals V2's residual failures are more than eight orders of magnitude more realistic.

II. Related Work

A. Adaptive Stress Testing

AST was originally introduced by Lee et al. [1] as a method for finding the most likely failure scenarios in black-box systems, initially applied to aircraft collision

avoidance using Monte Carlo Tree Search (MCTS). Subsequent work by Koren et al. [2] extended the AST framework to autonomous vehicles and introduced Deep Reinforcement Learning (DRL) policies as a highly scalable alternative to MCTS. The generalized AST framework, encompassing both MCTS and deep RL approaches, is formally detailed in [3].

B. Rare-Event Estimation via Importance Sampling

Importance sampling (IS) is a variance-reduction technique for estimating expectations under a target distribution by sampling from a different, more convenient proposal distribution [4]. In the context of safety validation, IS has been used to convert failure trajectories found under a biased search distribution back into probability estimates under the nominal environment distribution [5]. The combination of AST with IS was explored in [6], which surveys rare-event simulation methods for autonomous system validation.

C. Algorithmic Trading Safety

Technical analysis strategies such as SMA crossovers are among the most widely used systematic trading methods [7]. Robustness of such strategies to adversarial or anomalous market conditions is studied in the context of flash crashes [8] and adversarial market manipulation [9]. Our work bridges safety validation methodology and algorithmic trading, applying RL-based stress testing to estimate the probability of large drawdown events.

III. Background

A. Importance Sampling for Failure Probability

Let $\mathbf{x} \in \mathbb{R}^d$ be a vector of environmental disturbances drawn from a nominal distribution $p_0(\mathbf{x})$ and let $f(\mathbf{x}) \leq 0$ denote a failure event. The failure probability is

$$P_F = \mathbb{E}_{p_0}[\mathbf{1}\{f(\mathbf{x}) \leq 0\}] = \int \mathbf{1}\{f(\mathbf{x}) \leq 0\}\, p_0(\mathbf{x})\, d\mathbf{x}. \quad (1)$$

A naive Monte Carlo estimate $\hat{P}_F = \frac{1}{N}\sum_i \mathbf{1}\{f(\mathbf{x}_i) \leq 0\}$ has coefficient of variation $\sqrt{(1-P_F)/(NP_F)}$, which grows without bound as $P_F \to 0$ [4]. Importance sampling (IS) draws from a proposal $q(\mathbf{x})$ that concentrates mass near failures and corrects via likelihood ratios:

$$\hat{P}_F^{\text{IS}} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\{f(\mathbf{x}_i) \leq 0\} \cdot \frac{p_0(\mathbf{x}_i)}{q(\mathbf{x}_i)}, \quad \mathbf{x}_i \sim q. \quad (2)$$

This estimate is unbiased and has substantially lower variance when q is concentrated in the failure region [6].

B. Adaptive Stress Testing

Adaptive Stress Testing (AST) [1]–[3] frames failure-finding as an MDP $\langle \mathcal{S}, \mathcal{A}, T, R, \gamma \rangle$ where $\mathcal{S}$ is the joint system state, $\mathcal{A}$ is the space of environmental disturbances the adversary injects at each step, $T(s' \mid s, a)$ is the (black-box) system transition, and the reward $R(s, a, s')$ is designed to maximize failure probability. A reinforcement learning agent trained on this MDP learns a proposal policy π_θ that acts as an importance-weighted sampler over failure-inducing trajectories. Trajectory-level IS weights $w^{(i)} = \exp\!\big(\sum_t \log p_0(a_t) - \log \pi_\theta(a_t \mid s_t)\big)$ convert adversarial episodes into unbiased estimates of P_F under p_0.

IV. Methodology

A. Simulation Environment

We implement a custom Gymnasium environment, `TradingValidationEnv`, in which an adversary injects per-step noise ε_t into the market's log-return. The asset price evolves as:

$$p_{t+1} = p_t\,(1 + \mu + \sigma\varepsilon_t)\,, \quad \varepsilon_t \sim \mathcal{N}(0,1) \text{ (nominal)}, \quad (3)$$

where $\mu = 0.0001$ is the drift and $\sigma = 0.02$ is the volatility scale. The adversary's action space is $\varepsilon_t \in [-3, 3]$.

The five-dimensional observation at each step is:

$$s_t = [p_t,\ V_t,\ \text{SMA}_{50},\ \text{SMA}_{200},\ \mathbf{1}_{\text{invested}}]\,,$$

where SMA_{50} and SMA_{200} are the 50-step and 200-step simple moving averages of the asset price, V_t is the current portfolio value, and $\mathbf{1}_{\text{invested}}$ is a flag indicating whether the victim strategy is currently holding the asset.

A *failure* is defined as the portfolio value falling below $V_f = \$700$, corresponding to a 30% drawdown from the initial value of \$1,000. Each episode runs for at most 100 steps.

B. Victim Trading Strategies

Victim V1 (Baseline). V1 implements a standard SMA crossover strategy: the bot buys and holds the asset when $\text{SMA}_{50} > \text{SMA}_{200}$ and exits when $\text{SMA}_{50} \leq \text{SMA}_{200}$. To reflect realistic execution latency, an *exit lag* of 5 consecutive bearish steps is required before the position is liquidated. This lag creates exploitable inertia: the bot remains invested through the early phase of a crash.

Victim V2 (More Robust). V2 uses the same SMA-50/200 crossover logic as V1 but reduces the exit lag from 5 steps to 3. This directly addresses the exploit discovered in V1: by requiring only 3 consecutive bearish SMA steps before selling (instead of 5), the bot limits the window of inertia the adversary can exploit. The effect is an increase in *adversarial sample complexity*: an adversary trained for 200k timesteps exploits V1 trivially (100.0% failure), while doubling the training budget to 400k timesteps against V2 still yields only a $62.0 \pm 2.6\%$ failure rate, as shown in Table II.

C. Adversarial AST Agent

The adversary is a Proximal Policy Optimization (PPO) [10] agent with an MLP policy (two hidden layers of 64 units each), implemented using Stable-Baselines3 [11]. The reward signal is:

$$r_t = \underbrace{\frac{V_{t-1} - V_t}{2}}_{\text{damage reward}} \underbrace{-0.05\,\varepsilon_t^2}_{\text{likelihood penalty}} + \underbrace{1000 \cdot \mathbf{1}_{[V_t < 700]}}_{\text{crash bonus}}. \quad (4)$$

The likelihood penalty $-0.05\,\varepsilon_t^2$ is a soft constraint that penalizes actions far from the nominal $\mathcal{N}(0,1)$ distribution. This is essential for importance sampling validity: it ensures the adversary remains a proper proposal distribution rather than collapsing to a deterministic boundary policy. The two-component reward structure is intentional: the dense damage term $\frac{V_{t-1}-V_t}{2}$ provides a continuous gradient signal that steers the policy toward the failure region, while the sparse $+1000$ terminal bonus ensures that once the policy discovers a catastrophic drawdown, it anchors strongly on that behavior and does not trade it away for incremental damage rewards.

Adversary A1 is trained against V1 for 200,000 timesteps using four parallel environments, at which point training has converged. Adversary A2 is trained against V2 for 400,000 timesteps, double the budget of A1, to give the adversary ample opportunity to discover exploits against the more robust victim. The fact that A2 still achieves only $62.0 \pm 2.6\%$ failure rate despite twice the training compute directly quantifies the increase in adversarial sample complexity imposed by V2's reduced exit lag.

D. Monte Carlo Baseline

The MC baseline draws $\varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$ at each step with no learned policy, representing the natural stochastic environment. We run 500 MC episodes per victim version.

E. Importance Sampling Failure Probability Estimation

AST samples from the adversarial distribution π_θ, not the nominal distribution $\pi_0 = \mathcal{N}(0,1)$. To estimate $P_0(\text{failure})$ we apply trajectory-level importance sampling. For a trajectory of length T:

$$w^{(i)} = \exp\left(\sum_{t=1}^{T}\left[\log\pi_0\left(\varepsilon_t^{(i)}\right) - \log\pi_\theta\left(\varepsilon_t^{(i)} \mid s_t^{(i)}\right)\right]\right), \quad (5)$$

where $\log\pi_0(\varepsilon) = -\frac{1}{2}\varepsilon^2 - \frac{1}{2}\log(2\pi)$ and $\log\pi_\theta(\varepsilon \mid s)$ is obtained from the PPO policy via `evaluate_actions` in Stable-Baselines3 [11]. Two important considerations for IS validity. First, the likelihood penalty $-0.05\,\varepsilon_t^2$ in the reward (Eq. 3) penalizes $\mathbb{E}[\varepsilon_t^2] = \mu_t^2 + \sigma_t^2$, which encourages the policy to learn a *mean* closer to zero, not merely to reduce the magnitude of individual samples. These are distinct because only the former changes the IS denominator $\log\pi_\theta(\varepsilon_t \mid s_t)$ via the learned mean parameter. Crucially, because stochastic action sampling is used at evaluation time (not the deterministic mean), $\log\pi_\theta$ is evaluated under the exact same distribution from which actions are drawn, preserving IS validity regardless of how the penalty shapes the learned mean versus variance (see Sec. VI for further discussion). Second, `evaluate_actions` in SB3 returns log-probabilities under the unconstrained Gaussian $\mathcal{N}(\mu_t, \sigma_t^2)$, not the truncated distribution induced by clipping actions to the $[-3,3]$ interval. For V2, where the adversary's mean actions are moderate, this approximation is negligible. For V1, where actions frequently approach the ± 3 boundary, the pre-clipping log-prob can differ from the true truncated log-prob, introducing a systematic (though likely small) bias in IS weights. The IS estimate of the failure probability over N AST episodes is:

$$\hat{P}_0(\text{failure}) = \frac{1}{N}\sum_{i=1}^{N} w^{(i)} \cdot \mathbf{1}[\text{failure}_i]. \quad (6)$$

V. Experiments and Results

A. Experimental Setup

All experiments use the same environment reset distribution: the asset price is initialized uniformly at random from $[150, 200]$ with a 200-step upward price history, ensuring the victim starts in an invested (bull-market) state. All metrics are reported as mean $\pm$ standard deviation over 5 independent evaluation trials of 500 stochastic episodes each (2,500 total episodes per condition). Stochastic action sampling is used for AST so that importance weights are well-defined. Hyperparameters are summarized in Table I. Code to reproduce all experiments is available at https://github.com/EsawAdhana/ast-trading-validation.[1]

TABLE I
Hyperparameters

Parameter	**Value**
Environment	
Episode horizon T	100 steps
Action space ε_t	$[-3, 3]$
Failure threshold V_f	\$700 (30% drawdown)
Drift μ / volatility σ	0.0001 / 0.02
PPO Adversary (A1 and A2)	
Policy architecture	MLP, 2×64 hidden units
Learning rate	3×10^{-4}
Steps per rollout n_{steps}	2,048
Batch size	64
Entropy coefficient	0.01
Parallel environments	4
Training Budget	
A1 (vs. V1) total timesteps	200,000
A2 (vs. V2) total timesteps	400,000
Reward Weights	
Damage scale	1/2
Likelihood penalty λ	0.05
Crash bonus	1,000

The average negative log-likelihood (NLL) per episode is defined as:

$$\overline{\text{NLL}} = -\frac{1}{T}\sum_{t=1}^{T}\log\pi_0(\varepsilon_t) = \frac{1}{T}\sum_{t=1}^{T}\left[\tfrac{1}{2}\varepsilon_t^2 + \tfrac{1}{2}\log(2\pi)\right], \quad (7)$$

where $\pi_0 = \mathcal{N}(0,1)$ is the nominal distribution. Higher NLL indicates the episode's action sequence is less likely under the nominal market model; the theoretical minimum is $\frac{1}{2}\log(2\pi) \approx 0.919$ nats/step.

[1] All results can be regenerated by running `adversary.py` (training) followed by `multi_seed_eval.py` (evaluation).

B. Quantitative Results

Table II reports all five metrics across the four experimental conditions. Figure 1 visualizes representative adversarial and nominal trajectories, and Figure 2 shows the distribution of time-to-failure conditioned on failure episodes.

TABLE II
RESULTS: MEAN $\pm$ STD OVER 5 TRIALS $\times$ 500 EPISODES EACH

Metric	V1+AST	V1+MC	V2+AST	V2+MC
Fail Rate (%)	100.0 ± 0.0	0.4 ± 0.3	62.0 ± 2.6	0.4 ± 0.4
IS $\hat{P}_0$ (mean)	1.5×10^{-11}	N/A	2.0×10^{-3}	N/A
IS $\hat{P}_0$ (std)	7.7×10^{-12}	—	3.4×10^{-4}	—
NLL (nats/step)	4.868 ± 0.012	1.420 ± 0.005	1.561 ± 0.005	1.420 ± 0.005
Avg TTF (steps)	6.8 ± 0.0	51.5 ± 20.7	26.3 ± 0.3	55.5 ± 17.5
Avg Min Portf. ($)	677.5 ± 0.7	882.8 ± 2.9	705.2 ± 0.7	884.4 ± 2.8

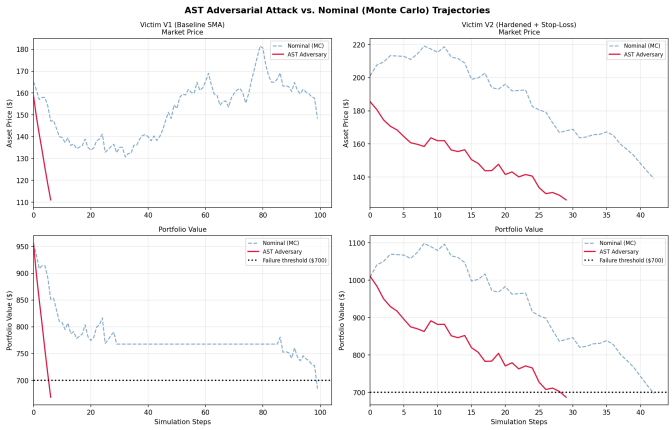

Fig. 1. Representative market price (top) and portfolio value (bottom) trajectories for adversarial AST (red) and nominal Monte Carlo (blue) conditions, for both the baseline V1 victim (left) and more robust V2 victim (right). The dashed horizontal line marks the $700 failure threshold.

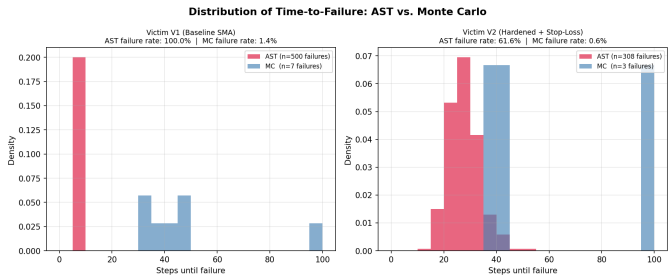

Fig. 2. Distribution of time-to-failure steps, conditioned on episodes where a failure occurred, for AST (red) and Monte Carlo (blue). V1+AST failures cluster sharply around 5–10 steps (mean 6.8 ± 0.0); V1+MC failures occur much later (mean 51.5 ± 20.7 steps) and are rare ($0.4 \pm 0.3\%$). V2+AST failures ($62.0 \pm 2.6\%$ rate) are spread across a wider range with a mean of 26.3 ± 0.3 steps, reflecting the longer, more nuanced attack sequence the adversary must discover against the more robust strategy.

C. Analysis

AST vs. Monte Carlo efficiency. Against V1, AST achieves $100.0 \pm 0.0\%$ failure rate across all trials, while MC achieves only $0.4 \pm 0.3\%$. This corresponds to a $\geq 250\times$ improvement in failure discovery efficiency. The difference in average time-to-failure is equally striking: AST induces failure in a mean of 6.8 ± 0.0 steps, while MC-found failures occur at a mean of 51.5 ± 20.7 steps. This indicates the adversary has learned a precise and quick-acting attack that exploits the V1 bot's 5-step exit lag: it delivers a rapid succession of large negative returns before the SMA crossover signal can trigger an exit. The zero standard deviation in V1+AST time-to-failure reflects the deterministic structure of the learned attack: A1 consistently triggers the 5-step exit lag condition within the same window regardless of the stochastic action draws.

Adversarial sample complexity of V2. Against the more robust V2 strategy, the adversary trained for twice the compute budget (400k vs. 200k timesteps) achieves only $62.0 \pm 2.6\%$ failure rate, compared to 100.0% against V1. The average time-to-failure nearly quadruples to 26.3 ± 0.3 steps, and the average minimum portfolio value ($705 vs. $678) is closer to the threshold, indicating the adversary is barely succeeding. This demonstrates that the 3-step exit lag forces the adversary to discover a longer, more nuanced attack sequence that requires significantly more training timesteps to reliably produce.

Importance sampling interpretation. The IS-corrected probabilities reveal a striking reversal of intuition. For V1, $\hat{P}_0(\text{failure}) \approx 1.5 \times 10^{-11}$ ($\pm 7.7 \times 10^{-12}$; see Limitations for further discussion): these failures require extremely non-nominal actions (avg NLL 4.868 ± 0.012 vs. 1.420 ± 0.005 for MC), corresponding to sustained $\approx 3\sigma$ market moves that are astronomically rare in practice. For V2, however, $\hat{P}_0(\text{failure}) \approx 2.0 \times 10^{-3}$ ($\pm 3.4 \times 10^{-4}$), more than eight orders of magnitude higher. V2's failures, while harder to find adversarially, occur through more moderate and realistic market sequences (avg NLL 1.561 ± 0.005, only slightly above the nominal 1.420 ± 0.005). This means that although V2 appears more robust by failure rate, its *found* failures are far more concerning from a real-world safety perspective. IS is essential here: a naive failure rate comparison (100.0% vs. 62.0%) would incorrectly suggest V2 is safer, while the IS-corrected picture is more nuanced.

VI. Discussion and Conclusion

We have demonstrated that Adaptive Stress Testing with importance-sampled failure probability estimation is an effective framework for the safety validation of algorithmic trading strategies.

The core advantage of AST over Monte Carlo is computational efficiency: AST finds failures $\geq 250\times$ more often than random sampling against V1 (100.0% vs. $0.4 \pm 0.3\%$), enabling rigorous characterization of failure modes within a practical compute budget. However, AST failure rate alone is an incomplete safety metric. V1 appears more dangerous by failure rate (100.0% vs. $62.0 \pm 2.6\%$), yet IS reveals V2's found failures are more than eight orders of magnitude more likely to occur naturally (2.0×10^{-3} vs.

1.5×10^{-11}). The IS step is therefore mandatory for converting adversarial findings into meaningful safety statements.

The iterative robustness improvement experiment illustrates a practical validation workflow: AST identifies the specific mechanism by which V1 fails (exploitation of exit lag via quick-acting crashes), which directly motivates the V2 design choice (reduced exit lag from 5 to 3 steps). The subsequent AST evaluation of V2 reveals both that V2 is harder to adversarially exploit and that its residual failures are far more realistic under the nominal distribution (2.0×10^{-3} vs. 1.5×10^{-11}). This pairing of adversarial sample complexity and IS-corrected probability is a more nuanced safety certificate than either metric alone could provide.

Limitations

The simulation model is simplified: asset price dynamics follow a single-asset, constant-volatility model, and the victim strategy does not incorporate transaction costs, slippage, or portfolio-level risk constraints. The 100-step episode horizon limits the adversary's ability to execute longer-horizon manipulation strategies. A further limitation concerns the nominal distribution itself: we model market noise as $\varepsilon_t \sim \mathcal{N}(0,1)$, which simplifies the importance sampling math but does not reflect the well-documented fat-tailed nature of real financial returns [12]. In practice, 3σ events occur far more frequently than a Gaussian predicts, meaning our IS-estimated probabilities may underestimate true real-world failure likelihoods for V1 and overstate the rarity of such events. Future work should replace $\mathcal{N}(0,1)$ with a heavier-tailed nominal distribution—such as a Student's t-distribution or a non-parametric kernel density estimate fitted to historical log-returns—to produce more calibrated failure probability estimates. Future work could also extend the framework to multi-asset portfolios, more realistic microstructure models, and victim strategies with learned (RL-based) policies, which would introduce a more complex adversarial interaction.

A further limitation concerns IS weight degeneracy for V1. The V1 adversary operates at a mean NLL of 4.868 nats/step versus the nominal 1.420, producing per-trajectory importance weights on the order of e^{-345} over a 100-step episode. Here, the effective sample size:

$$\mathrm{ESS} = \frac{\left(\sum_i w^{(i)}\right)^2}{\sum_i \left(w^{(i)}\right)^2}$$

is likely in the single digits, meaning the IS estimate of 1.5×10^{-11} rests on very few effective trajectories and should be interpreted as an order-of-magnitude indicator rather than a precise probability. The V2 estimate (2.0×10^{-3}, NLL gap of only 0.14 nats/step) does not suffer from this issue to the same degree.

A second IS validity caveat concerns action-space clipping. The adversary's action space is $\varepsilon_t \in [-3, 3]$, but the PPO policy is an unconstrained Gaussian; SB3 clips sampled actions to $[-3, 3]$ before stepping the environment. The log-probabilities returned by `evaluate_actions` are computed under the *pre-clipping* Gaussian, not under the truncated distribution that describes the actual sampling procedure. For V1, the adversary saturates near the ±3 boundary (avg. NLL 4.868 nats/step $\approx 3\sigma$ actions), so a non-trivial probability mass lies outside $[-3, 3]$ and the pre-clipping log-prob understates $\log \pi_\theta(\varepsilon_t \mid s_t)$, biasing IS weights upward. For V2, whose mean actions are far from the boundary (NLL 1.561 nats/step), the truncation effect is negligible. Correcting this bias would require replacing the Gaussian with an explicitly truncated normal or using a tanh-squashed policy, and is left for future work.

Conclusion. Safety validation of financial systems requires methods that can efficiently characterize rare but catastrophic failure events. AST, augmented with importance sampling, provides exactly this: efficient failure discovery combined with statistically valid probability estimation under the nominal operating distribution. The iterative robustness improvement loop of discovering failures, patching the system, and re-validating is a principled framework for improving robustness. Crucially, our results show that failure rate alone is an incomplete and potentially misleading safety certificate: V1 appears more dangerous by raw failure rate (100.0% vs. $62.0 \pm 2.6\%$), yet IS reveals V2's residual failures are more than eight orders of magnitude more likely to occur under natural market conditions (2.0×10^{-3} vs. 1.5×10^{-11}). The IS step is mandatory for converting adversarial findings into actionable safety statements about real-world risk.

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